

Chapter 2: Review of Probability and Statistics

R Exercises

1. Simulating a discrete random variable

The number of customer complaints W that a small company receives per day has the probability mass function

w_k	0	1	2	3	4
$p_W(w_k)$	0.10	0.25	0.30	0.20	0.15

- Compute $E(W) = \sum_k w_k p_W(w_k)$ and $\text{Var}(W) = \sum_k (w_k - E(W))^2 p_W(w_k)$ in R using the vectors `w <- 0:4` and `p <- c(0.10, 0.25, 0.30, 0.20, 0.15)`.
- Use `sample()` to simulate the number of complaints on $n = 1000$ days.
- Estimate the pmf by relative frequencies and compare with the true pmf. Draw a bar plot of the relative frequencies with `barplot()`.
- Compute the sample mean and the sample variance of your simulated data and compare them with the true values from (a).
- Repeat (b)–(d) with $n = 20$ days. What do you observe?

2. Densities in R

- The delay X (in hours) of a parcel delivery has the density $f_X(x) = x/2$ for $0 \leq x \leq 2$ and $f_X(x) = 0$ otherwise. Use the R function `integrate()` to check that f_X integrates to one, and to compute $P(0.5 \leq X \leq 1.5)$ and $E(X) = \int x f_X(x) dx$.
- Plot the density f_X with `curve()` for $x \in [-0.5, 2.5]$.
- Generate $n = 200$ realizations of a normal random variable with mean 1.6 and variance 2 using `rnorm()`.
- Estimate the density of your realizations with a histogram (`hist(..., freq = FALSE)`) and with a kernel density estimator (`density()`). Add the true density using `curve()` and `dnorm()`.
- Repeat (c) and (d) with $n = 20$ and with $n = 5000$. What do you observe?

3. Empirical cdf, sample moments, covariance and correlation

A retail chain records the advertising expenditures X (in 1000 EUR) and the sales Y (in 1000 units) of $n = 10$ of its stores:

$X_{i,\text{obs}}$	7	3	5	4	6	8	2	5	9	1
$Y_{i,\text{obs}}$	25	18	21	22	24	27	17	23	26	16

- Store the data in a data frame with the two columns `advertising` and `sales`.
- Compute the sample means, the sample variances and the sample standard deviations of both variables with `mean()`, `var()` and `sd()`.
- Compute the sample variance of X “by hand” in R, i.e., using only `sum()`, `length()` and `mean()`, and check that you get the same result as with `var()`.
- Compute and plot the empirical cdf of the sales with `ecdf()`. Use it to estimate the probability that a store sells more than 22,000 units.
- Draw a scatter plot of the data, and compute the sample covariance and the sample correlation with `cov()` and `cor()`.
- Convert advertising to EUR and sales to units, and recompute the covariance and the correlation. Explain the result.

4. The normal distribution in R

A machine fills coffee into cups. The filled amount X (in ml) is normally distributed with $\mu = 200$ and $\sigma = 5$.

- Use `pnorm()` to compute $P(X < 190)$ and $P(195 \leq X \leq 205)$.
- Use `qnorm()` to find the amount x that is undercut by only 5% of all cups.
- Standardize: compute $P(X < 190)$ a second time using `pnorm()` with its default arguments `mean = 0` and `sd = 1`.
- Simulate 10,000 fillings with `rnorm()`. Estimate $P(X < 190)$ and the 5% quantile from the simulated values (*hint*: `mean()` of a logical vector and `quantile()`), and compare with (a) and (b).
- Plot the densities of $\mathcal{N}(200, 5^2)$, $\mathcal{N}(210, 5^2)$ and $\mathcal{N}(200, 10^2)$ in one plot using `curve()` and `dnorm()`. How do μ and σ change the shape?

5. Chi-squared and F-distribution by simulation

Let $Z_1, \dots, Z_4$ be independent standard normal random variables and

$$W = Z_1^2 + Z_2^2 + \dots + Z_4^2 \sim \chi_4^2.$$

- Why can W never take negative values? What are $E(W)$ and $\text{Var}(W)$?
- Use R to generate 10,000 realizations of W *only* with the help of `rnorm()`.
Hint: `matrix(rnorm(10000 * 4), ncol = 4)` generates a matrix with 10,000 rows and 4 columns; `rowSums()` sums up each row.
- Compute the sample mean and the sample variance of your realizations and compare them with the values from (a).
- Draw a histogram of your realizations (`freq = FALSE`) and add the density of the χ_4^2 distribution using `curve()` and `dchisq()`.
- Estimate $P(W > 9.49)$ from your simulated realizations, and compare with the exact value from `pchisq()`.
- Now generate 10,000 realizations of

$$V = \frac{W_1/4}{W_2/20}, \quad W_1 \sim \chi_4^2, \quad W_2 \sim \chi_{20}^2, \quad W_1 \perp W_2,$$

using `rchisq()`. Draw a histogram and add the density of the $F_{4,20}$ distribution (`df()`).

6. Law of Large Numbers

Let D denote the result of rolling a fair die, $D \in \{1, 2, 3, 4, 5, 6\}$, with $E(D) = 3.5$.

- Use R to simulate $n = 1000$ rolls of a fair die with `sample()`.
- Compute the running sample means

$$\bar{D}_{m,\text{obs}} = \frac{1}{m} \sum_{i=1}^m D_{i,\text{obs}} \quad \text{for } m = 1, 2, \dots, 1000,$$

and plot them against m . Add a horizontal line at $E(D) = 3.5$.

Hint: `cumsum(d) / seq_along(d)` computes all running means at once.

- Repeat (a) and (b) a few times. What changes from repetition to repetition, and what does not?
- Now consider a loaded die with $P(D = 6) = 0.5$ and $P(D = k) = 0.1$ for $k = 1, \dots, 5$. Compute $E(D)$ for this die by hand. Repeat (a) and (b) for the loaded die and check your result.

7. Simulating from a data-generating process

The monthly electricity costs Y_i (in EUR) and the living space X_i (in m^2) of households are generated by the DGP

$$Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

with $\beta_1 = 30$, $\beta_2 = 0.8$, $\sigma = 15$, and $X_i \sim \mathcal{N}(80, 20^2)$ independent of ε_i .

- Simulate $n = 200$ households from this DGP.
- Draw a scatter plot and add the true regression line $\beta_1 + \beta_2 x$ with `abline()`.
- Split the simulated households into three groups: $X_i \leq 60$, $60 < X_i \leq 100$ and $X_i > 100$. Compute the mean of X_i and the mean of Y_i within each group and add these three points to your plot. What do you observe?
- Compute the sample correlation between X_i and Y_i . Then repeat the simulation with $\sigma = 40$. How do the scatter plot and the correlation change?

8. Simulating a confounder

Consider again the DGP for ice cream sales X_i and swimming pool visitors Y_i , which are both driven by the daily maximum temperature Z_i :

$$\begin{aligned} Z_i &\sim \mathcal{N}(22, 5^2), \\ X_i &= 2 + 0.5 Z_i + u_i, \quad u_i \sim \mathcal{N}(0, 1), \\ Y_i &= 1 + 0.3 Z_i + v_i, \quad v_i \sim \mathcal{N}(0, 1), \end{aligned}$$

with Z_i , u_i and v_i independent of each other.

- Simulate $n = 3650$ days (ten years) from this DGP.
- Draw a scatter plot of (X_i, Y_i) and compute the sample correlation.
- Now only keep the days with a maximum temperature between 24 and 26 degrees Celsius. How many days are left? Compute the sample correlation between X_i and Y_i for these days and compare with (b).
- Color the points in your scatter plot from (b) by temperature group ($Z_i < 18$, $18 \leq Z_i \leq 26$, $Z_i > 26$). What do you see?